# MAJI NDOGO

## Water Access & Community Safety Report

Visualising the Currents of Change

## Executive Summary

This report presents a data-driven analysis of water access and community safety in Maji Ndogo, a fictional nation of approximately 27.6 million people. Drawing on field survey records, AI-enhanced queue photography, and a decade of crime data linked to water-collection sites, the findings reveal a deep crisis — one that disproportionately burdens women and children.

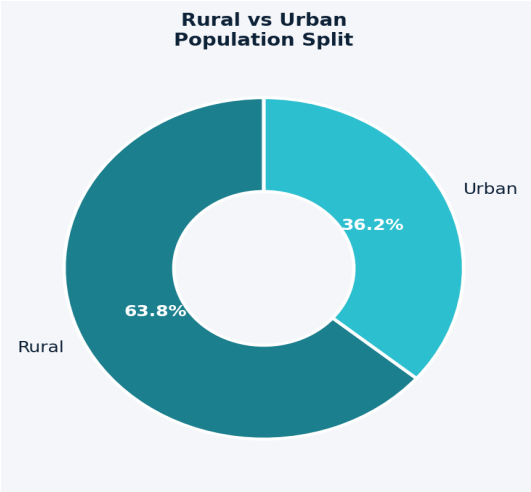
| 27.6M | 63.8% | 43.2% | 13.8% |
|---|---|---|---|
| Total Population | Live in Rural Areas | Use Shared Taps | Broken Home Taps |

**Key Finding: Women and children bear the heaviest burden — they constitute the majority of those queuing at shared taps and represent the largest share of crime victims near water-collection points.**

# 1. Water Access Overview

## 1.1 Population Distribution

Nearly two-thirds of Maji Ndogo's population resides in rural areas, where access to clean, safe water is most severely constrained. Urban residents benefit from a proportionally higher share of functioning tap infrastructure.



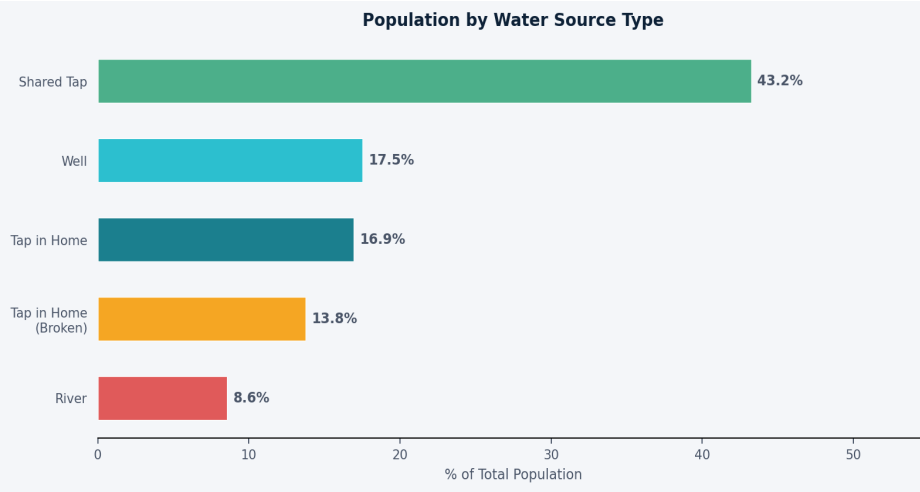




Fig 1a: Rural/Urban split (left) · Fig 1b: Water source breakdown (right)

## 1.2 Key Water Source Insights

- **Shared taps** serve 43.2% of the population — by far the most common source, with up to 2,000 people sharing a single tap.
- **Tap-in-home** infrastructure reaches 16.9%, but an additional 13.8% have **broken** home-tap systems that urgently need repair.
- **Wells** serve 17.5% of the population; however only an estimated 28% of wells are classified as clean.
- **Rivers** remain the only option for 8.6% of citizens — the least safe and least reliable water source.

## 1.3 Provincial Tap-in-Home Coverage

The map below (created in Power BI using MD_Provinces.json) allows province-level filtering. The table shows the count of functioning tap-in-home sources per province.

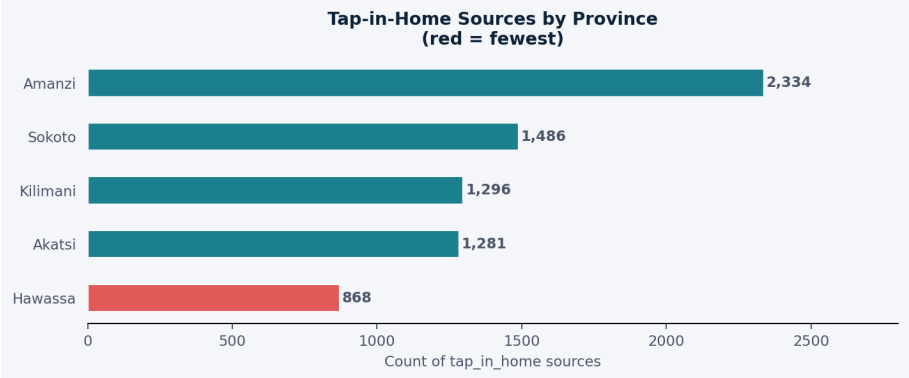



Fig 2: Tap-in-home sources by province — Hawassa has the fewest (868).

**Key Finding: Hawassa has only 868 tap-in-home sources, the lowest of all provinces, making it the top priority for new infrastructure investment.**

| Province | Tap-in-Home Sources | Priority |
|---|---|---|
| Hawassa | 868 | ■ Highest |
| Akatsi | 1,281 | ■ High |
| Kilimani | 1,296 | ■ Medium |
| Sokoto | 1,486 | ■ Medium |
| Amanzi | 2,334 | ■ Lower |

# 2. Queue Times & Burden of Collection

## 2.1 Daily Queue Patterns

Citizens of Maji Ndogo face an average wait of over 120 minutes to collect water. Queue times vary dramatically by day — Saturday queues are nearly three times longer than those on Sunday or Wednesday, the shortest-queue days.

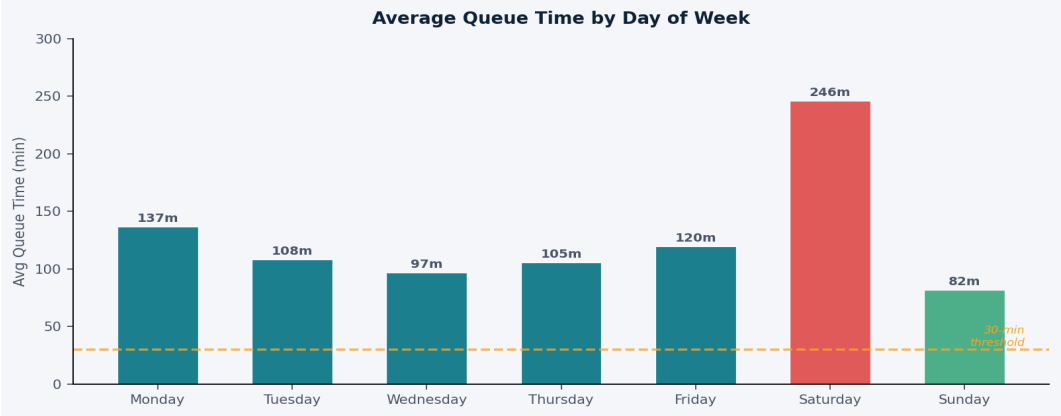



Fig 3: Average queue time by day of week. The dashed line marks the 30-min policy threshold.

**Key Finding: Saturday queues average 246 minutes — over 4 hours. Residents who must collect water should target Wednesday or Sunday to minimise waiting time.**

## 2.2 Gender Composition of Queues

Analysis of AI-tagged queue photographs reveals a stark gender imbalance. Women dominate the queues on weekdays, while the gender split becomes more balanced on weekends. This suggests that full-time employment constrains men to weekend collection, leaving women to shoulder the daily burden.

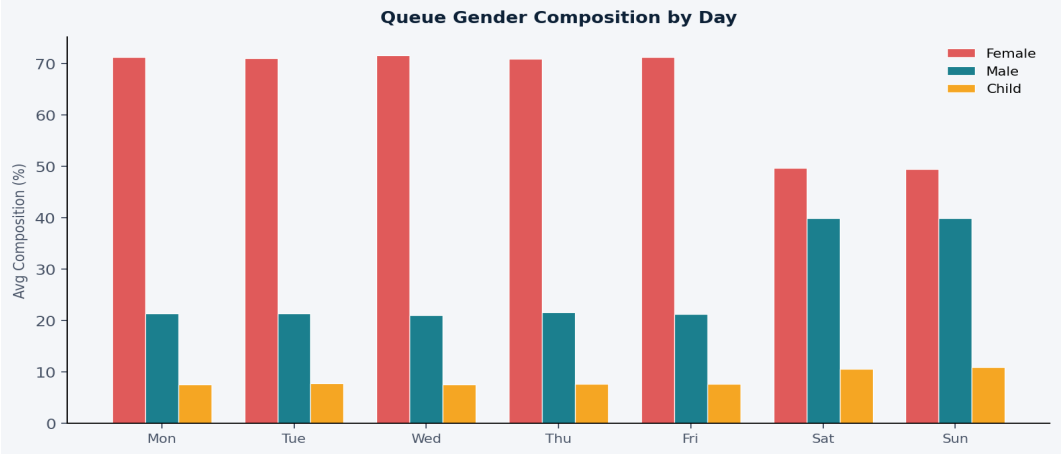



Fig 4: Average gender composition per day. Women consistently make up the majority.

## Key patterns observed:

- Women form the **majority of queue members on every weekday**, often exceeding 60–70% of the queue.
- On **weekends, the gender split equalises** significantly — consistent with men having more leisure time.
- Children are present in queues across all days, reflecting that **mothers bring children** along while collecting water.
- Towns with the highest proportion of children in queues at shared taps include **Kintampo and Lusaka**.
- Provinces with the best gender equality in water access: **Amanzi** shows the most balanced male/female queue composition.

# 3. Crime Linked to Water Collection

A decade of crime records (77,226 incidents) have been geo-linked to water-source locations. Crimes occur at these sites, or during travel to and from them. Women and children represent **74.8%** of all victims.

## 3.1 Crime Type Breakdown

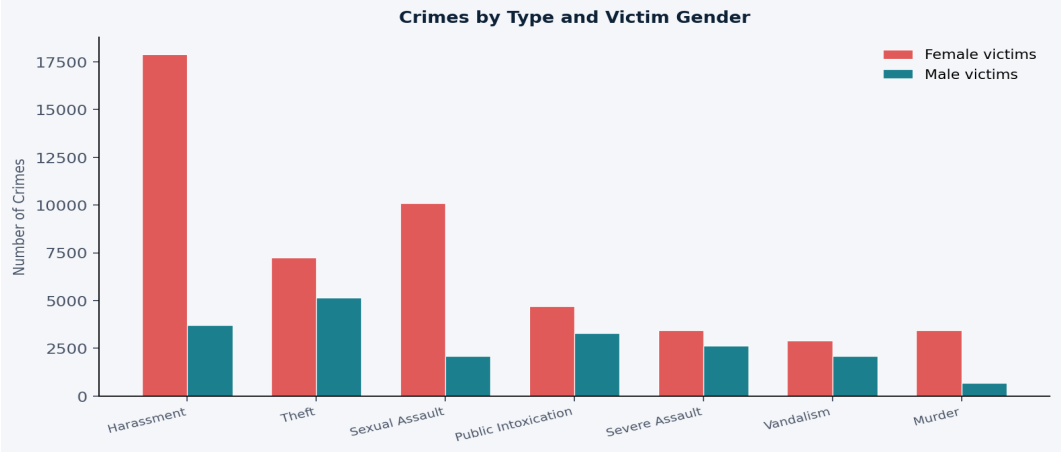



Fig 5: Crime count by type, split by victim gender. Harassment is the most prevalent crime.

| Crime Type | Total | Female Victims | % Female |
|---|---|---|---|
| Harassment | 23,104 | 17,904 | 77.5% |
| Theft | 14,431 | 7,243 | 50.2% |
| Sexual Assault | 13,065 | 10,102 | 77.3% |
| Public Intoxication | 9,404 | 4,696 | 49.9% |
| Severe Assault | 7,093 | 3,434 | 48.4% |
| Vandalism | 5,753 | 2,893 | 50.3% |
| Murder | 4,376 | 3,452 | 78.9% |

**Key Finding: Harassment is the most reported crime with 23,104 incidents — more than twice the rate of theft (14,431) or sexual assault (13,065). Women are the primary victims across all crime categories.**

## 3.2 Provincial Crime Distribution

Crime levels vary considerably across provinces. Amanzi stands out as having the lowest number of crimes against women — making it the safest province for female water collectors. Kilimani and Akatsi record the highest female victim counts.

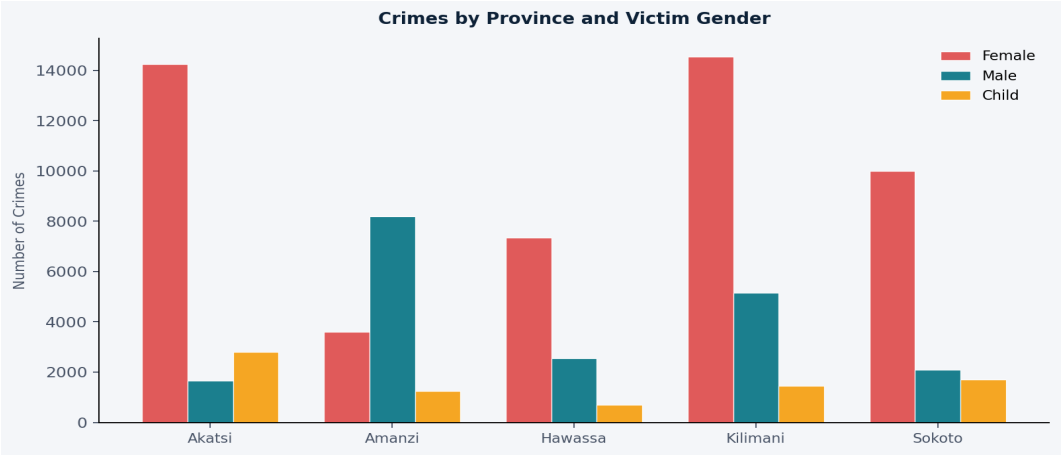



Fig 6: Crime counts by province and victim gender.

## 4. Recommendations

| | |
|---|---|
| Infrastructure | • Install additional shared taps in provinces with the highest queue times, prioritising sources serving >500 people. Adding a second tap near a 3,000-person shared tap can halve queue times from ~200 min to ~100 min. |
| Infrastructure | • Repair broken tap-in-home systems urgently in Hawassa (868 sources) — the province with the lowest functional tap coverage and highest unmet need. |
| Gender Equity | • Deploy community water stewards to accompany collectors during high-risk hours (early morning, late evening) to reduce harassment and assault incidents. |
| Gender Equity | • Introduce water-collection scheduling systems to distribute load more evenly across days, reducing Saturday peak queues and associated safety risks. |
| Safety | • Focus law enforcement resources near water points in Kilimani and Akatsi, which record the highest female victim counts across all crime types. |
| Safety | • Establish well-lit, visible queue areas at shared taps to deter crime — particularly in provinces with high harassment rates near water sites. |
| Data | • Continue AI-assisted queue photography analysis to track changes in gender composition over time as infrastructure improvements are implemented. |

### Data Sources

• Md_summary.csv — Field survey data including water source type, location, queue times, number of people served, and AI-derived gender composition of queues.

• Md_queue_related_crime.csv — Crime records (2010–2020) geo-linked to water source locations, including crime type, victim gender, and time of incident.

• MD_Provinces.json — TopoJSON map file defining the five provincial boundaries of Maji Ndogo.